

Problem 1

$((c) \implies (a))$

Assume (c)

Let $\varphi : G \rightarrow H \times K$ s.t. $\varphi(g) = (\phi'(g), \psi(g))$

Homomorphism:

Since ϕ' and ψ are homomorphisms φ is a homomorphism.

Injective:

Let $g \in \ker(\varphi)$

Thus $\varphi(g) = (\phi'(g), \psi(g)) = (e, e)$ thus $g \in \ker(\phi'), \ker(\psi)$ thus since this is an exact sequence $\ker(\psi) = \text{im}(\phi)$ thus since $\phi' \circ \phi = \text{id}_H \implies g = e$ thus $\ker(\varphi) = \{e\}$

Surjective: Let $(h, k) \in H \times K$, thus since σ is surjective there exists a $g \in G$ s.t. $\psi(g) = k$ and for the h consider $\phi(h)$

Thus $\varphi(\phi(h), g) = (\phi'(\phi(h)), \psi(g)) = (h, k)$ thus surjective. Thus $G \cong H \times K$ thus proven in class since this is a direct product this sequence is split.

$((a) \implies (b))$

Assume (a)

Thus shown in class this means there exists a $K' \cong K$ s.t. $G = \phi(H)K$ and $\phi(H) \cap K' = 1$

Consider $\psi|_{K'} : K' \rightarrow K$ is a homomorphism.

Injective:

Let $k' \in \ker(\psi|_{K'})$ thus $\psi|_{K'}(k') = e \in \phi(H) \implies k' = e$ since $\phi(H) \cap K' = \{e\}$

Surjective:

For any $k \in K$, since ψ is surjective, there exists $g \in G$ s.t. $\psi(g) = k$

Thus you can write $g = \phi(h)k'$ for $h \in H$ and $k' \in K'$

$k = \psi(g) = \psi(\phi(h)k') = \psi(\phi(h))\psi(k') = \psi(k')$

Thus $\psi|_{K'}$ is surjective.

Since $\psi|_{K'}$ is an isomorphism take its inverse and define $\psi' : K \rightarrow G$ s.t. $\psi' = (\psi|_{K'})^{-1}$ we can do this since $K' \subset G$

ψ' is a homomorphism since $(\psi|_{K'})^{-1}$ is a homomorphism.

Consider $\psi(\psi'(k))$ for $k \in K$ well as shown above ψ' is actually only going to map to K' and because of this we can think of $\psi = \psi|_{K'}$ thus since these are inverse of each other

$\psi(\psi'(k)) = k$ thus done

$((b) \implies (a))$

Assume (b)

Define $K' = \psi'(K)$

Let $x \in \phi(H) \cap K'$

Thus $x = \phi(h) = \psi'(k)$ for $h \in H$ and $k \in K$

$\psi(\phi(h)) = \psi(\psi'(k)) \implies e = k \implies x = \psi'(e) = e$ thus $\phi(H) \cap K' = \{e\}$

Let $g \in G$

Define $k' = \psi'(\psi(g)) \in K'$

Consider $\psi(gk'^{-1}) = \psi(g)\psi(k'^{-1}) = \psi(g)\psi(k')^{-1} = \psi(g)\psi(\psi'(\psi(g)))^{-1} = \psi(g)\psi(g)^{-1} = e$

Thus $gk'^{-1} \in \ker(\psi) = \text{im}(\phi)$ thus there is an $h \in H$ s.t. $\phi(h) = gk'^{-1}$ thus $g = \phi(h)k'$

Thus $G = \phi(H)K'$

Thus G is split

Problem 2

For a group of order 8 to be split in this sequence $G \cong C_2 \rtimes C_2 \times C_2$ thus there exists a $\phi : C_2 \times C_2 \rightarrow \text{Aut}(C_2) \cong \{e\}$ thus the only homomorphism $\phi(a) = \text{id}_{C_2 \times C_2}$ s.t. $a \in C_2$ thus to be split $G \cong C_2 \times C_2 \times C_2$

1. C_8

If $N \cong C_2$, then $C_8/N \cong C_4$ thus C_8 cannot fit in this sequence.

2. $C_4 \times C_2$

$$C_2 \cong \langle (a^2, e) \rangle$$

$$C_4 \times C_2 / \langle (a^2, e) \rangle = \{ \langle (a^2, e) \rangle, (a, e) \langle (a^2, e) \rangle, (e, b) \langle (a^2, e) \rangle, (a, e) \langle (a^2, e) \rangle \} \cong C_2 \times C_2$$

$$e \rightarrow \langle (a^2, e) \rangle \rightarrow G \rightarrow G / \langle (a^2, e) \rangle \rightarrow e$$

Thus this sequence fits and is not split from above.

3. $C_2 \times C_2 \times C_2$

$$C_2 \cong \langle (a, e, e) \rangle$$

$$C_2 \times C_2 \times C_2 / \langle (a, e, e) \rangle = \{ \langle (a, e, e) \rangle, (e, b, e) \langle (a, e, e) \rangle, (e, e, c) \langle (a, e, e) \rangle, (e, b, c) \langle (a, e, e) \rangle \} \cong$$

$$C_2 \times C_2$$

$$e \rightarrow \langle (a, e, e) \rangle \rightarrow G \rightarrow G / \langle (a, e, e) \rangle \rightarrow e$$

Thus this sequence fits and is split by above.

4. D_4

$$C_2 \cong \langle x^2 \rangle$$

$$D_4 / \langle x^2 \rangle = \{ \langle x^2 \rangle, x \langle x^2 \rangle, y \langle x^2 \rangle, xy \langle x^2 \rangle \} \cong C_2 \times C_2$$

$$e \rightarrow \langle x^2 \rangle \rightarrow G \rightarrow G / \langle x^2 \rangle \rightarrow e$$

Thus this sequence fits and is not split from above.

5. Q

$$C_2 \cong \langle -1 \rangle$$

$$Q / \langle -1 \rangle = \{ \langle -1 \rangle, i \langle -1 \rangle, j \langle -1 \rangle, k \langle -1 \rangle \} \cong C_2 \times C_2$$

Thus this sequence fits and is not split from above.

Problem 3

If G is split, there must exist a subgroup $D'_2 \cong D_2$ in G . This implies we can center all reflections and rotations at a single point, as there is a subgroup isomorphic to D_2 .

Let v be the smallest vector on the x -axis, and let w be the vector with the smallest height. Let z be the smallest vector in the lattice.

- If z lies on the x -axis, then $z = v$.
- If z lies on the y -axis, then $z = w$.

Assuming without loss of generality that z is in the first quadrant, we can act on it by r , so $r(z)$ is in our lattice. Then $r(z) + z$ lies on the x -axis. Since z is the smallest vector in the lattice, we have:

$$x(r(z) + z) < 2x(v) \Rightarrow 2x(z) < 2x(v)$$

If $2x(w) \neq x(v), 0$, then $z + r(z) - v$ would be a smaller vector than v on the x -axis, which is a contradiction.

Thus, if $x(z) = 0$, then $z = w$. If $x(z) = \frac{1}{2}x(v)$, then by similar reasoning, we find $x(z) = x(w)$. Since w has the smallest height and z is the smallest vector, it must be w . Therefore, our lattice is spanned by v and w .

We now have two cases: whether w is vertical or not.

1. **If w is vertical**, this corresponds to case 9:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_2 = \langle x, y, r, p_\pi \mid xy = yx, rp_\pi = p_\pi r, r^2 = e, p_\pi^2 = e, rxr = x, ryr = y^{-1}, p_\pi x p_\pi = x^{-1}, p_\pi y p_\pi = y^{-1} \rangle$$

with

$$\phi(r) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_\pi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

2. **If w is not vertical**, this corresponds to case 12:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_2 = \langle x, y, r, p_\pi \mid xy = yx, rp_\pi = p_\pi r, r^2 = e, p_\pi^2 = e, rxr = x, ryr = xy^{-1}, p_\pi x p_\pi = x^{-1}, p_\pi y p_\pi = y^{-1} \rangle$$

with

$$\phi(r) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_\pi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

If G is not split, then we have two cases:

1. If one reflection lifts to only a glide.

2. If both reflections lift to only a glide.

For the first case, if one reflection lifts to a glide, we make the glide parallel to the x -axis and the reflection parallel to the y -axis, centering the rotations at the origin. Let the min glide (meaning moves the minimum distance) parallel to the x -axis be defined as $t_{(a,h)}rt_{(0,-h)} = t_{(a,2h)}r$. Let a reflection parallel to the y -axis be $t_{(b,0)}r_y t_{(-b,0)} = t_{(2b,0)}r_y$.

Now, consider:

$$t_{(a,2h)}rt_{(a,2h)}r = t_{(a,2h)}t_{(a,-2h)}rr = t_{(2a,0)}$$

Thus, we have a vector on the x -axis. We call the minimal vector on the x -axis v . Assume $v = (c, 0)$ with $c < a$. Then consider:

$$t_{(-c,0)}t_{(a,2h)}r = t_{(a-c,2h)}r$$

Since $a - c < a$, this contradicts the minimal glide assumption, so $v = (2a, 0)$.

Similarly, we can find a vector on the y -axis by acting with r_y , and then finding the minimal vector, which we call w .

Let z be the smallest vector. If z lies on any axis, it must be v or w . If the x - or y -coordinate of z is zero, then it is either v or w , leaving us with $z = (a, 0.5y(w))$ using a similar argument as above.

Consider the element:

$$t_{-z}t_{(a,2h)}r = t_{(0,2h-0.5y(w))}r$$

Then:

$$t_{(0,2h-0.5y(w))}rt_{(0,2h-0.5y(w))}r = t_{(0,2h-0.5y(w))}t_{(0,-2h+0.5y(w))}rr = \text{id}$$

This is a reflection parallel to the horizontal axis, which is not allowed. Thus, this minimal vector z cannot exist. Therefore, the minimal vector is either v or w , so these two span the lattice.

Thus:

$$t_{(a,2h)}r = t_{\frac{1}{2}v+cw}r \in G \quad \text{for } c \in \mathbb{R}$$

and

$$t_{(2b,0)}r_y = t_{kv}r_y \in G \quad \text{for } k \in \mathbb{R}$$

Now, consider:

$$t_{kv}r_y t_{\frac{1}{2}v+cw}r = t_{(k-\frac{1}{2})v+cw}p\pi \in G$$

Thus:

$$t_{(k-\frac{1}{2})v+cw}p\pi p\pi = t_{(k-\frac{1}{2})v+cw} \in G$$

implying $k - \frac{1}{2} \in \mathbb{Z}$ and $c \in \mathbb{Z}$ since this translation is in my group, and we only have integer linear combinations of v and w .

So:

$$t_{-cw}t_{\frac{1}{2}v+cw}r = t_{\frac{1}{2}v}r = t_{(a,0)}r \in G$$

Also, since $k - \frac{1}{2} \in \mathbb{Z}$ and $2b = 2ak \Rightarrow b = ak \Rightarrow k = \frac{d}{2}$ for d odd, we have $b = \frac{ad}{2}$. Setting $d = 1$, we find $b = \frac{a}{2}$, so:

$$t_{(a,0)}r_y \in G$$

Thus, we have case 11:

$$G = \langle \mu, y, r, p\pi \mid \mu^2 y = y\mu^2, r^2 = p\pi^2 = e, r\mu r = \mu^{-1}, rp\pi = p\pi r, ry r = y, \mu y \mu^{-1} = \mu r, p\pi \mu^{-1} = \mu p\pi, p\pi \mu p\pi = \mu^{-1}, p\pi y p\pi = y^{-1} \rangle$$

If two reflections lift to glides, we pick the minimal glides $t_{(a,2c)}r$ and $t_{(2d,b)}r_y$ parallel to their respective axes. The minimal vectors on the x -axis and y -axis, as in the previous proof, are $v = (2a, 0)$ and $w = (0, 2b)$, respectively.

Then:

$$t_{(a,2c)}r = t_{\frac{1}{2}v+kw}r \quad \text{for } k \in \mathbb{R}$$

$$t_{(2d,b)}r_y = t_{jv+\frac{1}{2}w}r_y \quad \text{for } j \in \mathbb{R}$$

Now consider:

$$t_{jv+\frac{1}{2}w}r_y t_{\frac{1}{2}v+kw}r = t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w}p\pi \in G$$

Thus:

$$t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w}p\pi p\pi = t_{(\frac{1}{2}-j)v+(\frac{1}{2}+k)w} \in G$$

so we require $\frac{1}{2} - j \in \mathbb{Z}$ and $\frac{1}{2} + k \in \mathbb{Z}$.

Since $k + \frac{1}{2} \in \mathbb{Z}$ and $2c = 2ak$, it follows that $k = \frac{f}{2}$ for some odd integer f . Choosing $f = 1$, we get $c = \frac{a}{2}$. Similarly, $d = \frac{b}{2}$, so $t_{(a,0)}r \in G$ and $t_{(0,b)}r_y \in G$.

Thus, we have case 10:

$$G = \langle \mu, \gamma, p\pi \mid \mu^2 \gamma^2 = \gamma^2 \mu^2, \gamma \mu^{-1} = \mu \gamma^{-1}, \mu p\pi \mu^{-1} = \mu^2 p\pi, \mu \gamma^{-1} = \gamma \mu^{-1}, \gamma p\pi \gamma^{-1} = \gamma^2 p\pi \rangle$$

Problem 4

If G lifts one reflection to a reflection s.t. a rotation of order 4 is on that line; thus, make the reflection like the x -axis, let this reflection be r with the center of the rotation $p_{\pi/2}$ at the origin.

Let x be a vector. We can act on x by r to obtain something on the x -axis. Let v be the minimal vector on the x -axis.

Applying a rotation, we obtain a vector on the y -axis. Let w be the smallest vector on the y -axis.

Now, consider $p_{\pi/2}(v) - w$; this vector would be shorter than w , which is a contradiction. Thus, $p_{\pi/2}(v) = w$, meaning they must be the same vector.

Let z be the smallest vector and, without loss of generality, assume it lies in the first quadrant. By a similar argument as above, we either conclude that z is on one of the axes or that $z = (0.5 \cdot x(v), 0.5 \cdot x(v))$.

If this is our minimal vector, we can rotate by 45° to get a new coordinate system where the lattice is spanned by z and $p_{\pi/2}(z)$ or by v and w .

Thus, for any coord. system we will have a line of reflection with a center of an order 4 rotation.

Then, we can use $p_{\pi/2}$ on the reflection to get all reflections

Thus, this is a clear subgroup in G that is isomorphic to D_4 thus we are in the split case

Thus, with a square lattice we have case 13:

$$G \cong \mathbb{Z} \times \mathbb{Z} \rtimes D_4 = \langle x, y, r, p_{\pi/2} \mid xy = yx, rp_{\pi/2} = p_{\pi/2}^{-1}r, r^2 = e, p_{\pi/2}^4 = e, rxr = x, ry r = y^{-1}, p_{\pi/2} x p_{\pi/2}^{-1} = y, p_{\pi/2} y p_{\pi/2}^{-1} = x^{-1} \rangle$$

For:

$$\phi(r) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \phi(p_{\pi/2}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

If G does not lift any reflection to a reflection such that a rotation of order 4 lies on that line, then this is the non-split case. We cannot have four reflections intersecting at a rotation point of order 4, which would be necessary for a subgroup isomorphic to D_4 .

Thus, at least one reflection lifts only to a glide, let it be parallel to the x -axis, and place the center of an order-4 rotation at the origin. Let $t_{(a,2h)}r$ be the minimal glide. As shown in previous proofs, the minimal vector on the x -axis is $(2a, 0) = v$. Similarly, by acting on this vector with a 90° rotation, we find that the minimal vector on the y -axis is $(0, 2a) = w$.

Now, consider the minimal vector z . By previous reasoning, z must be v , w , or $0.5(v + w)$. If $z = 0.5(v + w)$, then let our basis for the lattice be z and $p_{\pi/4}(z) = w$.

Then:

$$t_{-z}t_{(a,2h)}r = t_{(0,2h-a)}r = t_{kw}r \in G \quad \text{for } k \in \mathbb{R}$$

This gives:

$$t_{kw}rt_{kw}r = \text{id} \in G$$

which implies a reflection parallel to the x -axis.

Since $p_{\pi/4}t_{kw}r = t_{-kz}r_{y=-x} \in G$,

$$t_{-kz}r_{y=-x}t_{-kz}r_{y=-x} = t_{-kz}t_{kw} = t_{-kz+kw}$$

which implies $k \in \mathbb{Z}$, as this translation is in G .

Thus, $t_{-kw}t_{kw}r = r \in G$, which is a contradiction because we assumed there were no reflections through a center of a rotation of order 4. Therefore, v and w must span our lattice.

Now:

$$t_{(a,2h)}r = t_{\frac{1}{2}v+kw}r \quad \text{for } k \in \mathbb{R}$$

and

$$p_{\pi/4}t_{\frac{1}{2}v+kw}r = t_{\frac{1}{2}w-kv}r_{y=-x} \in G$$

Then:

$$t_{\frac{1}{2}w-kv}r_{y=-x}t_{\frac{1}{2}w-kv}r_{y=-x} = t_{\frac{1}{2}w-kv}t_{-\frac{1}{2}v+kw} = t_{-(k+\frac{1}{2})v+(\frac{1}{2}+k)w} \in G$$

which implies $k + \frac{1}{2} \in \mathbb{Z}$, so $k = \frac{d}{2}$ for d odd. Choosing $d = 1$, we find $2ka = 2h \Rightarrow h = \frac{a}{2}$.

Thus, $t_{(a,a)}r$ is our minimal glide.

Consider:

$$p_{\pi/4}^2 t_{(a,a)}r = t_{(-a,-a)}r_y \in G$$

so the inverse of this, $t_{(-a,a)}r_y$, is also in G . This is the minimal glide parallel to the y -axis; otherwise, we would contradict the minimality of w on the y -axis.

Additionally, we can obtain the minimal glides on the lines $y = x$ and $y = -x$ by rotating the glide parallel to the x -axis:

$$p_{\pi/4}t_{(a,a)}r = t_{(-a,a)}r_{y=-x}$$

$$p_{\pi/4}^3 t_{(a,a)}r = t_{(a,-a)}r_{y=x}$$

These are minimal along their lines, as shown by similar arguments as before. We can also get the reflections by adding v or w to these glides:

$$t_{(a,a)}r_{y=-x} \in G$$

$$t_{(a,a)}r_{y=x} \in G$$

and

$$t_{(a,a)}r_{y=-x}t_{(a,a)}r_{y=-x} = \text{id}$$

$$t_{(a,a)}r_{y=x}t_{(a,a)}r_{y=x} = \text{id}$$

Thus, this is the only other group representation with two glides in the vertical and horizontal directions, and two reflections along the lines $y = x$ and $y = -x$, giving us case 14 for a square lattice:

$$G \cong \langle \mu, p_{\pi/2} \mid \mu^2 p_{\pi/2} \mu^2 p_{\pi/2}^3 = p_{\pi/2} \mu^2 p_{\pi/2}^3 \mu^2, p_{\pi/2} \mu p_{\pi/2}^{-1} = \mu^{-1}, p_{\pi/2} \mu^{-1} = \mu p_{\pi/2}^{-1} \rangle$$

Problem 5

Theorem 0.0.1. *An isomorphism between wallpaper groups takes translations to translations, rotations to rotations, reflections to reflections and glide reflections to glide reflections.*

Proof. Let $\varphi : G \rightarrow H$ be an isomorphism between wallpaper groups.

Let $t \in G$ be a translation. Since translations have infinite order $\varphi(t)$ is either a glide or a translation.

Assume for contradiction that $\varphi(t)$ is a glide. Let y be a translation that is not parallel to the glide. Thus, these do not commute.

Since φ is an isomorphism it is surjective thus there exists a $g \in G$ s.t. $\varphi(g) = y$. Thus, again g is either a glide or translation, but we know g^2 is a translation in either case.

However, this implies y^2 and $\varphi(t)$ commute since $\varphi(g^2t) = \varphi(tg^2)$. Thus, this means that y^2 is parallel to the glide φ which is a contradiction since y is not parallel.

Thus, translations go to translations and glides go to glides.

Let $r \in G$ be a reflection. Since reflections have order 2 $\varphi(r)$ is either a reflection or a 180 turn.

Let $t \in G$ be a translation not perpendicular to the reflection line of r or equivalent s.t. tr is a glide.

Thus, $\varphi(tr) = \varphi(t)\varphi(r)$ this is a product of a translation and a 180 turn which is still a 180 turn. Thus, this is a contradiction from the first part where glides go to glides.

Thus, reflections go to reflections.

Thus, finally rotations are forced to go to rotations. □

Corollary 0.0.2. *If two wallpaper groups are isomorphic then their point groups are isomorphic.*

Proof. Let $G \cong H$ for two wallpaper groups. With translations T, T' and point groups K, K' for G and H respectively.

Since they are isomorphic this means there exists an isomorphism φ between them.

Thus, by the previous theorem we get that $\varphi(T) = T'$ thus $T \cong T'$

Thus, $K \cong G/T \cong H/T' \cong K'$ □

Now using the corollary we can say if two wallpaper groups have different point groups they are not isomorphic thus we can split it up into groups.

1. For all the point groups of C_n there is only one type for each number, so we are done with those.

2. For point group D_1

We proved in class these are not isomorphic.

3. For point group D_2

First using the theorem since case 9,12,11 have reflections and case 10 does not we know that case 10 is not isomorphic to 9,12,11.

For, case 11 prove above since at any center of a rotation of order 2 we only have one reflection this makes it, so we will not have a subgroup isomorphic to D_2 thus 11 is not isomorphic to 9 and 12.

Let G be case 9, and H be case 12.

I tried.

4. For point group D_3

I tried.

5. For point group D_4

Let G be our case 13 and H be our case 14.

Assume for contradiction $G \cong H$ with $\varphi : G \rightarrow H$.

First in G center ourselves at a center of rotation of order 4, and let one reflection be the x -axis.

Thus, we have that $rr_{y=x} = p_{\pi/2}$. Thus, $\varphi(rr_{y=x}) = \varphi(r)\varphi(r_{y=x}) = \varphi(p_{\pi/2})$

Thus, by the theorem since we assumed these are isomorphic we have that two reflections in H can produce a rotation of order 4.

Now in H center ourselves at a center of rotation of order 4. Then set a basis v, w for our lattice. Let v be horizontal and w be vertical.

Thus shown in a previous problem the only reflections we can get in H are parallel to the line $y = x$ and $y = -x$.

Now lets try to get to $t_c p_{\pi/2} = t_z r_a t_x r_b$ for z, x, c as vectors and a, b as $y = x$ or $y = -x$

$t_c p_{\pi/2} = t_{z+r_a(x)} r_a r_b = t_{z+r_a(x)} p_{j\pi/2} p_{i\pi/2} = t_{z+r_a(x)} p_{(i+j)\pi/2}$ for $i, j \in \{1, 3\}$ however there is no combination of i, j to get to $p_{\pi/2}$ thus this is a contradiction. Thus, G is not isomorphic to H .

6. For point group D_6

There is only one group with point group D_6 , so we are done.